

ASSIGNMENT 1

PNAP0143: Logical Reasoning

ASASIpintar

Universiti Kebangsaan Malaysia

Session 2026/2027

[100 marks]

Instruction: Students are required to form groups of two to three members from the same set to complete the following assignment. Please ensure that all answers demonstrate detailed steps and full solutions. Submissions must be strictly handwritten in pen on A4 size paper (including foolscap); printed versions of digital handwriting generated via tablets will not be accepted. Please attach the provided cover page to your assignment and submit the physical hard copy to your lecturer on or before **5:00 p.m. on Friday, 14 August 2026**.

1. For any integers a and b , $a + b \geq 15$ implies that $a \geq 8$ or $b \geq 8$. Prove the statement by using:
 - a. direct proof,
 - b. proof by contrapositive method.
2. Consider the bi-conditional statement:

"For any integer n , $3n + 2$ is an even integer if and only if n is an even integer."

 - a. Briefly explain which proof methods would be most suitable to prove both directions ($p \rightarrow q$ and $q \rightarrow p$) of this bi-conditional statement.
 - b. Hence, show the proof in (a).
3. Prove by contradiction that if a and b are integers and $a^2 + b^2$ is divisible by 4, then both a and b must be even integers.
4. Use the Principle of Mathematical Induction to prove that for all positive integers $n \in \mathbb{N}$:

 $4^n - 1$ is divisible by 3

5. Given that

$$\frac{2x^3 + 7x^2 + 2x - 3}{x^2 - 5x + 3} \leq 0$$

Use long division to factorise the cubic expression. Hence, solve for x and write your answer in interval notation.

6. Solve for x :

$$|x + 5| < |4x - 6| - 2$$

7. Let $f(x) = \sqrt{x}$.

- Sketch the graph of $g(x) = -2\sqrt{x+3} + 1$ using transformation rules. Describe each transformation step (reflection, shifting, and stretching/compressing).
- Determine the domain and range of $g(x)$.

8. Calculate the following limits using Limit Laws. Show all algebraic manipulations:

a. $\lim_{x \rightarrow \infty} \frac{5x^2 + 8x - 3}{3x^2 + 2}$

b. $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 5} - 3}{x - 2}$

c. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 6x} - x)$

9. Given the function $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$:

- Identify the horizontal and vertical asymptotes of $f(x)$ by calculating the appropriate limits.
- Justify why a specific x -value is a vertical asymptote and evaluate the one-sided limits to explain the behavior of the graph at any other x -values where the function is initially undefined.

10. Use the Precise Definition of a Limit ($\epsilon - \delta$ definition) to prove that:

$$\lim_{x \rightarrow 2} (5x - 3) = 7$$

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Set	
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Member's Name	Matric Number

Lecturer

	Ms. Nur Shamim binti Hamzah
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